

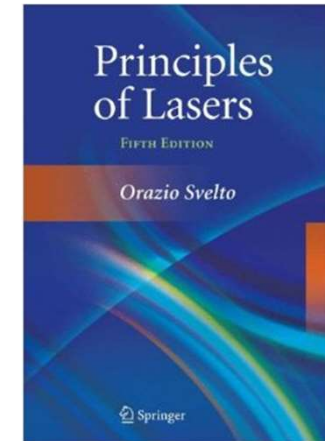


南方科技大学
SOUTHERN UNIVERSITY OF SCIENCE AND TECHNOLOGY

《激光原理与技术》

Principles and Technologies of Lasers

Ch4-2 Oscillation Characteristics of Lasers



Dr. Dan Luo(罗丹)

Email: luod@sustech.edu.cn

Office: Room 331 at College
of Engineering South Tower

Tel: 0755-88018552



Chapter 4: Oscillation Characteristics

❖ 4.2 Oscillation Mode (Output Mode)

- Mode competition in lasers with homogenous broadening
- Multi-longitudinal mode oscillation of lasers with inhomogeneous broadening

❖ 4.3 Output Optical Power and Energy

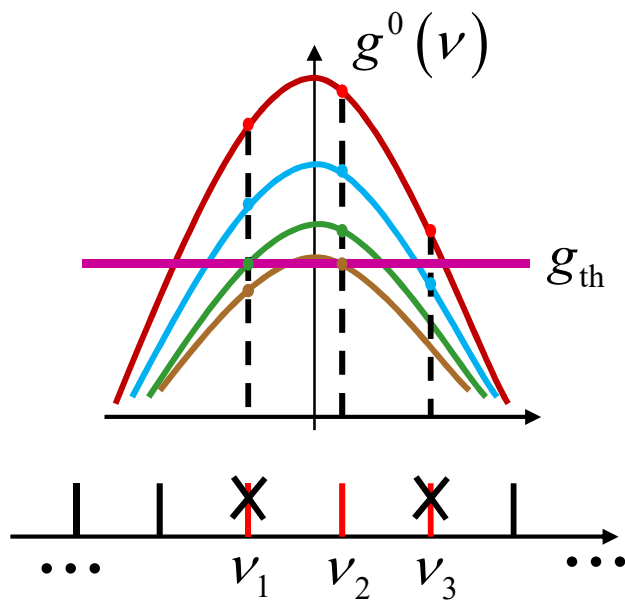
- Output power for CW or long-pulse laser
- Output power of short-pulse lasers ($t_0 \ll \tau_s$)

4.2 Oscillation mode (output mode)

Discuss the output modes of lasers from the mechanism of gain saturation

❖ Mode competition in lasers with homogenous broadening

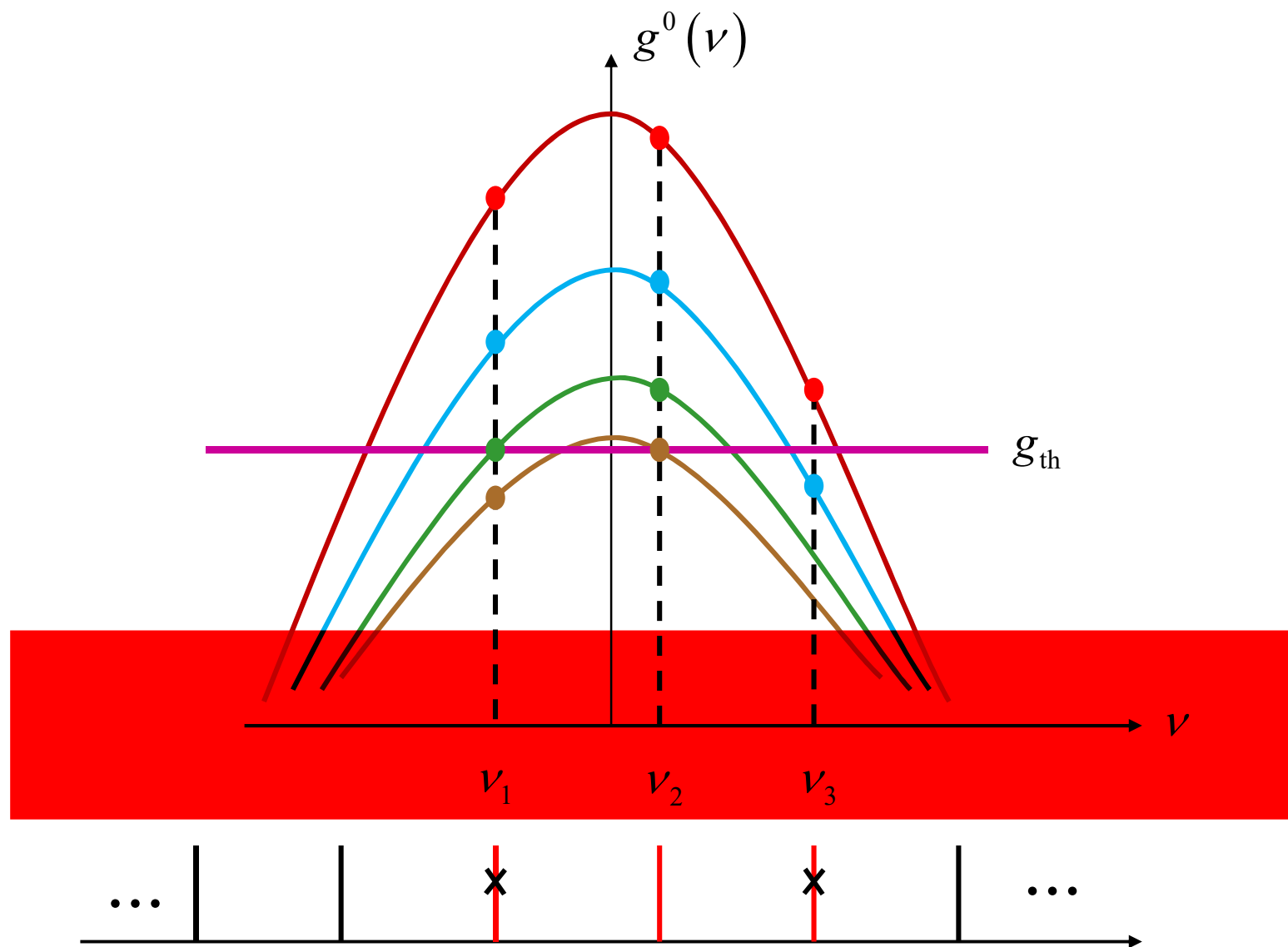
❑ Self-selection mode due to homogenous saturation of gain



Gain Curve	ν_1		ν_2		ν_3	
$g(\nu)$	$g(\nu_1)$	I_1	$g(\nu_2)$	I_2	$g(\nu_3)$	I_3
Red	$> g_{th}$	$\uparrow$	$> g_{th}$	$\uparrow$	$> g_{th}$	$\uparrow$
Blue	$> g_{th}$	$\uparrow$	$> g_{th}$	$\uparrow$	$= g_{th}$	-
Green	$= g_{th}$	-	$> g_{th}$	$\uparrow$	$< g_{th}$	0
Brown	$< g_{th}$	0	$= g_{th}$	-	$< g_{th}$	0

steady-state gain

Conclusion: the output of a homogeneous broaden steady-state laser should be a single longitudinal mode. Its working frequency locates at central frequency of the gain curve. Other longitudinal modes will be suppressed.



□ Multimode oscillation induced by spatial hole-burning

1) Spatial hole burning effect

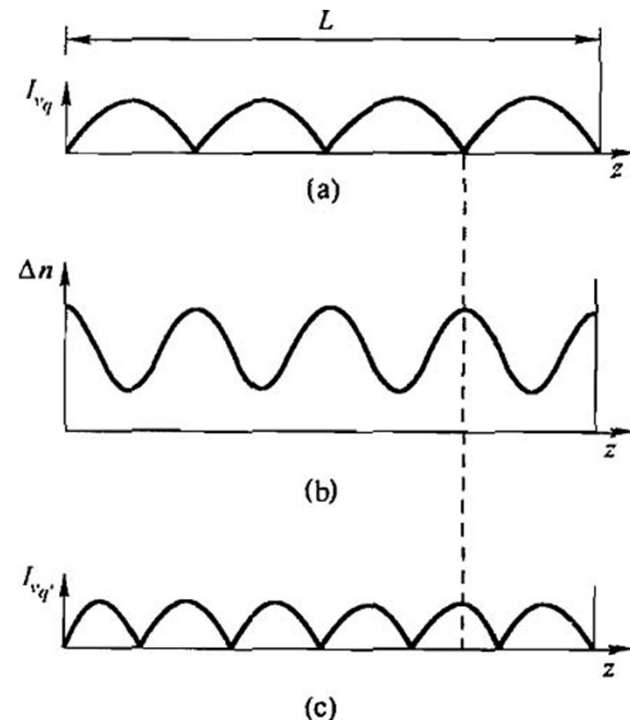
- Assume homogeneous transverse distribution, only consider z -direction distribution
- Standing wave $\Rightarrow$ gain distribution $g(z) \Rightarrow$ spatial hole-burning

2) Physical mechanism

- Different longitudinal modes can use the activated atoms in different positions inside the cavity

3) Formation conditions

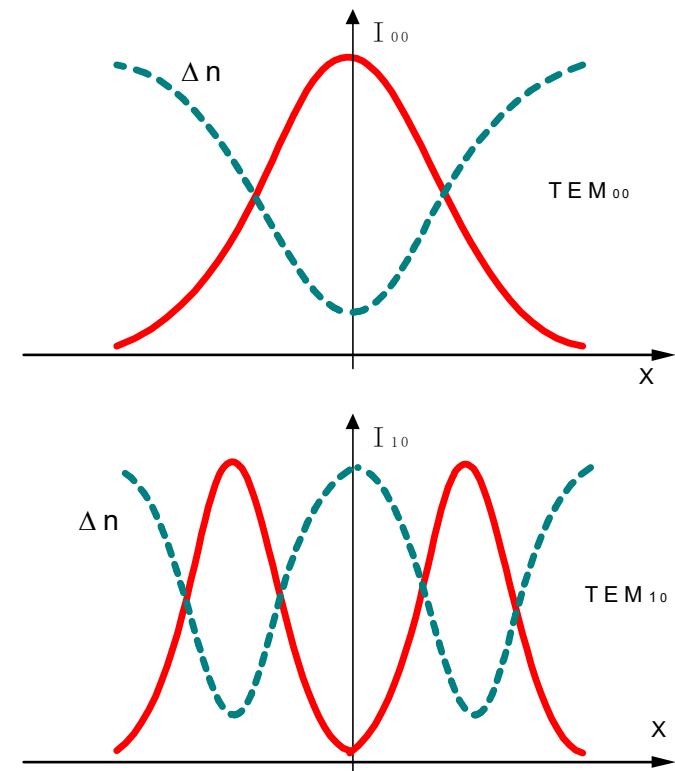
- Standing wave cavity: spacing between burning holes is at the scale of wavelength
- Slow spatial transfer speed of activated atoms



- Gas: random thermal motion, fast spatial transfer speed, difficult to form spatial hole-burning.
- Solid: example, Cr ions bound to the lattice structure, $\lambda/4$ spatial transfer needs 10^{-4}s
- Semiconductor: 10^{-7}s

4) Formation mechanism of transverse hole-burning

- Nonuniform distribution of population of transverse modes, transverse hole-burning size is large (mm scale), spatial transfer cannot eliminate the nonuniformity.
- With enough pumping, different transverse modes can utilize the activated atoms in different positions and hence form multi-transverse modes oscillation.

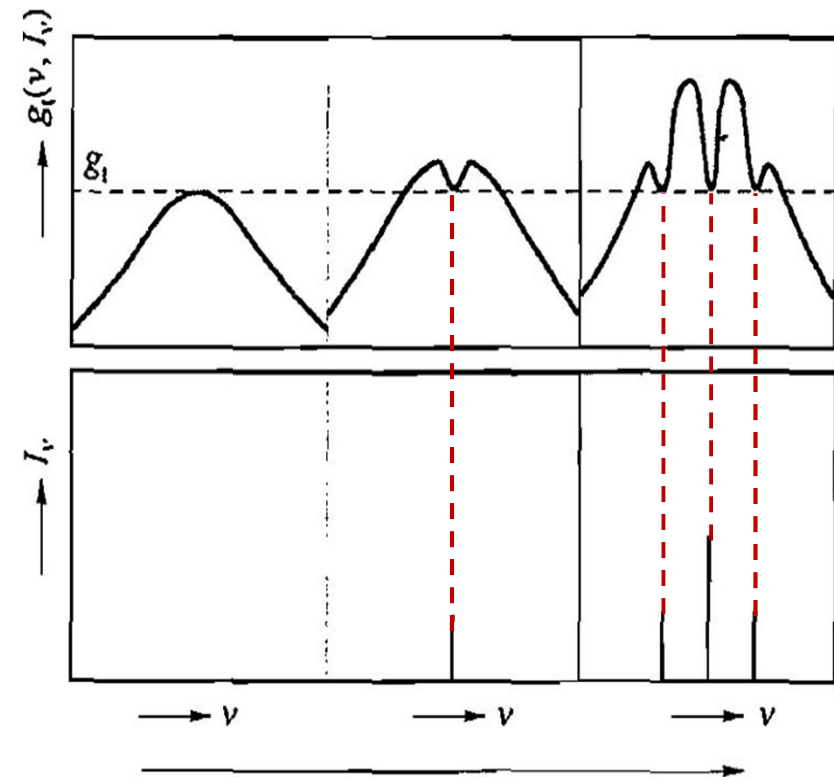


How about for pulse excitation? Mode competition time is in the range of μs

❖ Multi-longitudinal mode oscillation of lasers with inhomogeneous broadening

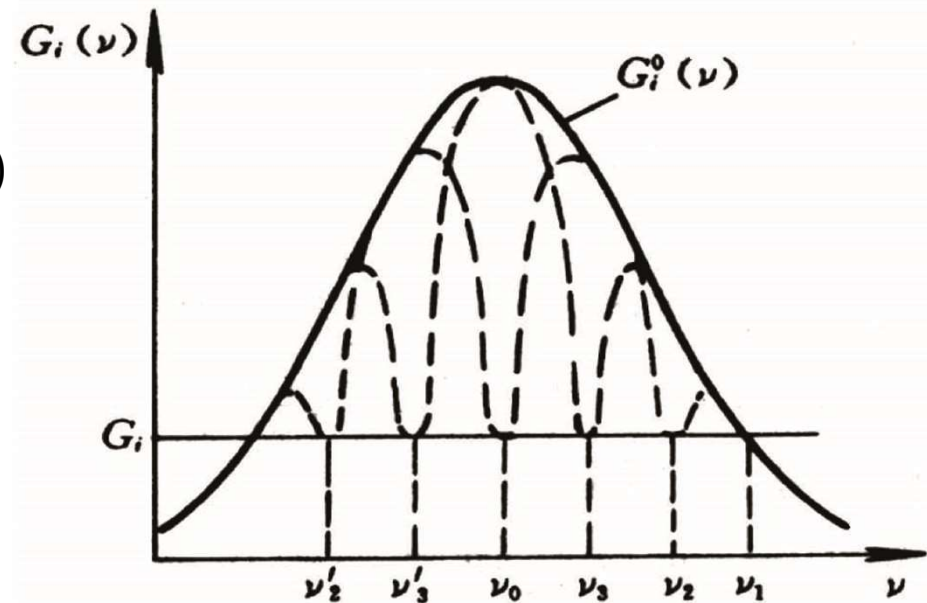
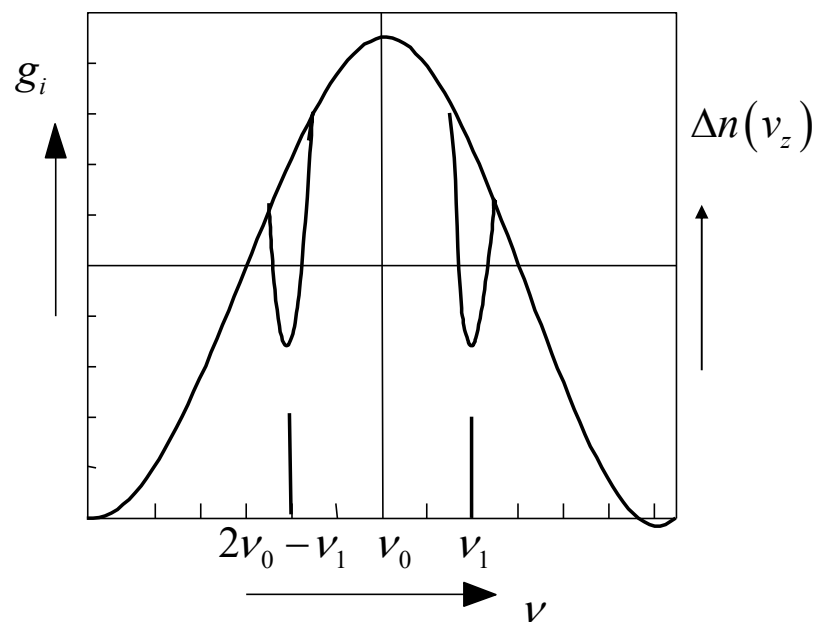
External pump $\uparrow \rightarrow g^0 \uparrow \rightarrow$ longitudinal modes that meet threshold conditions $\uparrow \rightarrow$ mode numbers $\uparrow$

- For incident light (ν, I_ν) , gain saturation will only happen in the frequency range of $\nu + d\nu$. Laser oscillation can still happen for other frequencies.
- For the case of Doppler broadening, symmetrical hole burning at both sides of central frequency ν_0 . If $\nu_q = \nu_0$, modes ν_{q+1} and ν_{q-1} will compete, and result in random fluctuation of their output power
- If frequency spacing between ν_q and ν_{q+1} is less than the spacing of burning hole width, their burning holes will overlap and result in mode competition



□ Mode competition in inhomogeneous broadened lasers with standing wave

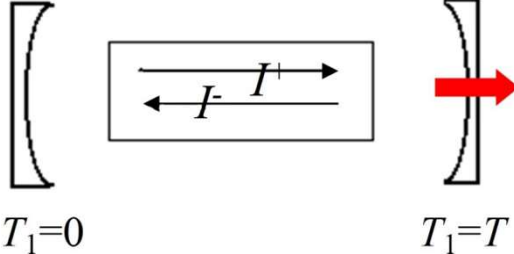
- Longitudinal mode frequencies ν_1 and ν_2 symmetrically distribute on both sides of the central frequency ν_0 , and deplete the inverted population
- The spatial burning holes will overlap for adjacent longitudinal modes if the spacing is small enough



4.3 Output Optical Power and Energy

Discuss the average intensity at steady-state, estimate the output power

Area of the laser beam's cross-section

$$P = I^+ TA \Rightarrow P = \frac{1}{2} I TA \Rightarrow P_{out} \Leftrightarrow I^+$$


❖ Output power for CW or long-pulsed laser

▣ Single-mode laser (Assume mode l with frequency of ν_q)

$$\begin{cases} \Delta n^0 > \Delta n_t \\ g^0(\nu_q) > g_t \end{cases} \Rightarrow \begin{cases} N_l \uparrow \\ I_{\nu_q} \uparrow \end{cases} \xrightarrow{\text{saturation}} \begin{cases} \Delta n \downarrow \\ g(\nu_q, I_{\nu_q}) \downarrow \end{cases} \xrightarrow{g(\nu) > g_t} \begin{cases} N_l \uparrow \\ I_{\nu_q} \uparrow \end{cases} \xrightarrow{\text{enhanced saturation}} \begin{cases} \Delta n \downarrow \\ g(\nu_q, I_{\nu_q}) \downarrow \end{cases}$$

$$\begin{cases} g(\nu_q, I_{\nu_q}) = g_t = \alpha \\ \frac{dN_l}{dt} = 0, \quad \frac{d\Delta n}{dt} = 0 \end{cases} \Rightarrow \begin{matrix} I_{\nu_q} & \text{fixed} \\ & \text{intensity} \\ N_l & \text{no change} \end{matrix}$$

$$N_l \nu h \nu = I_{\nu_q}^+ + I_{\nu_q}^- \Rightarrow I_{\nu_q}^+ = \frac{1}{2} N_l \nu h \nu$$

$$P = I_{\nu} TA = \frac{1}{2} N_l h \nu \nu AT$$

□ Lasers with homogeneous broaden ($\nu = \nu_0$)

Assume $T \ll 1$, $I_\nu^+ \approx I_\nu^-$, $I = I_\nu^+ + I_\nu^- \approx 2I_\nu$

How to calculate I_ν inside cavity

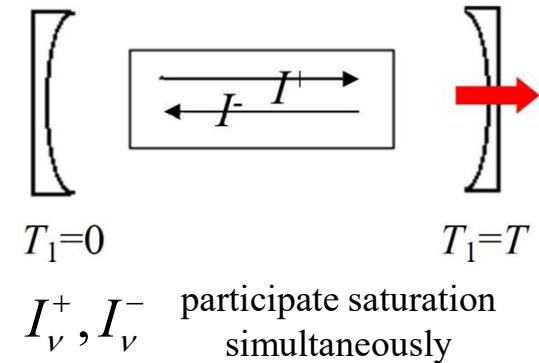
$$g_H(\nu, I_\nu) = \frac{g_m}{1 + (2I_\nu/I_s)} = \frac{\delta}{l} = g_t$$



$$I_\nu = \frac{1}{2} I_s \left(\frac{g_m l}{\delta} - 1 \right)$$



$$P = \frac{1}{2} I_s T A \left(\frac{g_m l}{\delta} - 1 \right)$$



Gas lasers: $T \ll 1$

$$2\delta \approx T + \alpha$$

$\alpha \ll 1$

$$P = \frac{1}{2} I_s T A \left(\frac{2g_m l}{\alpha + T} - 1 \right)$$

- α — single-loop net loss factor without considering reflection
- A — effective sectional area of the laser beam

Discussion 1: Output power $\sim$ pumping power ($P_p \uparrow \rightarrow P \uparrow$)

➤ Solid-state lasers (optical pumping)

$$P = \frac{1}{2} I_s T A \left(\frac{g_m l}{\delta} - 1 \right)$$

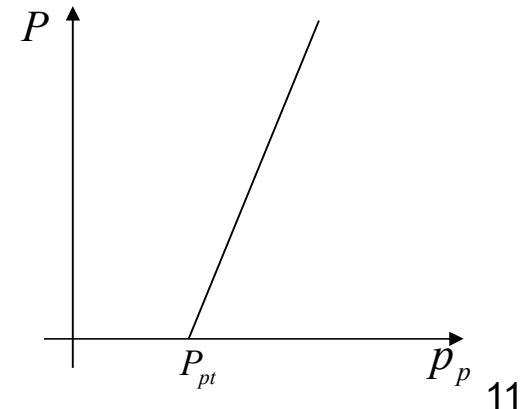
$$\left. \begin{aligned} g_m &= \Delta n^0 \sigma_{21} & P_p &= \frac{h\nu_p \Delta n^0 V}{\eta_F \tau_s} \\ g_t &= \Delta n_t \sigma_{21} & P_{pt} &= \frac{h\nu_p \Delta n_t V}{\eta_F \tau_s} \end{aligned} \right\} \begin{aligned} &\eta_2 = \frac{\tau_2}{\tau_s} \quad \eta_F = \eta_1 \eta_2 \\ &\frac{\Delta n_t}{\tau_2} = \frac{P_{pt} \eta_1}{h\nu_p V} = \frac{P_{pt} \eta_1}{h\nu_p S l} \end{aligned}$$

$$P = \frac{1}{2} I_s T A \left(\frac{g_m l}{\delta} - 1 \right) = \frac{1}{2} h\nu_0 \frac{\Delta n_t l}{\delta \tau_2} T A \left(\frac{P_p}{P_{pt}} - 1 \right) = \frac{A \nu_0}{S \nu_p} \eta_0 \eta_1 P_{pt} \left(\frac{P_p}{P_{pt}} - 1 \right)$$

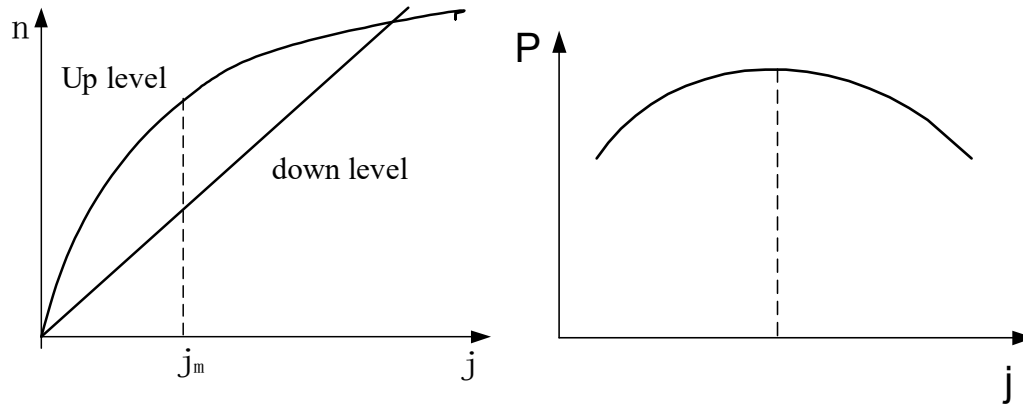
$$I_s = \frac{h\nu_0}{\sigma_{21} \tau_2}, \quad \frac{\delta}{l} = \Delta n_t \sigma_{21} \quad \eta_0 = T/2\delta$$

When $P_p > P_{pt}$, P increases linearly with P_p

- The output power of the optically pumped laser is converted from the pump power beyond P_{pt}
- I_ν is determined by the gain beyond g_t



➤ Gas lasers: exist optimum discharge current (P229)



Empirical formula of the optimal discharge current for He-Ne laser

$$g_m = 1.4 \times 10^{-2} \frac{l}{d}$$

➤ Semiconductor lasers (P278)

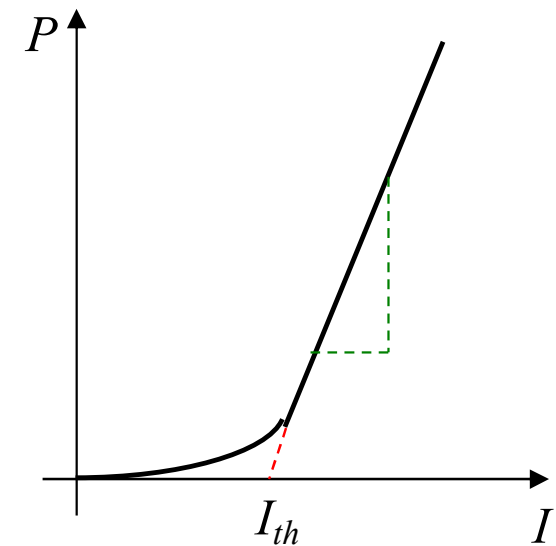
$$P = \frac{J - J_{th}}{e} \eta_i \hbar \omega \quad P_{out} = \frac{\ln(1/r)}{\alpha L + \ln(1/r)} P$$

$$\eta_D = \frac{P_{out}}{P_{in}} = \frac{J - J_{th}}{J} \eta_i \frac{\hbar \omega}{eV} \frac{\ln(1/r)}{\alpha L + \ln(1/r)}$$

➤ η_i – Carrier recombination probability

➤ η_D – LD (laser diode) efficiency

➤ $P_{in} = JV$



Discussion 2: Output power ~ transmittance (optimum T)

➤ Can be Experimentally measured and calculated

$$T \uparrow \quad \delta \uparrow \rightarrow g_t \uparrow \rightarrow I_v \downarrow \quad \Rightarrow \quad T_m \quad \boxed{P = \frac{1}{2} I_s T A \left(\frac{g_m l}{\delta} - 1 \right) = \frac{1}{2} I_s T A \left(\frac{2g_m l}{a + T} - 1 \right)}$$

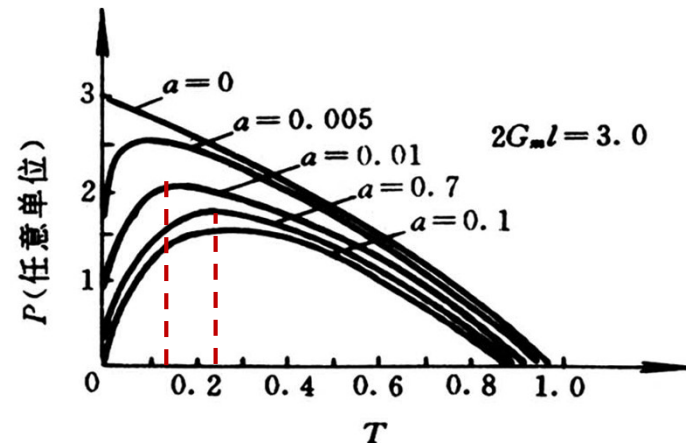
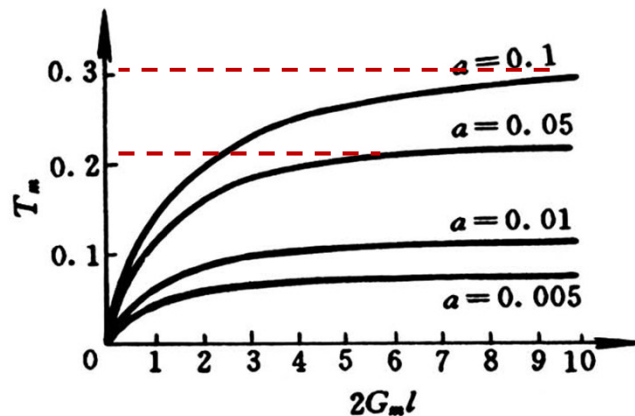
T_m : 1) Experimentally fixed P_p , change T and measure P_{out}

2) Calculate $dP/dT=0$

$$P = \frac{1}{2} I_s T A \left(\frac{2g_m l}{a + T} - 1 \right) \quad \Rightarrow \quad \frac{dP}{dT} = A I_s g_m l \frac{a}{(a + T)^2} - \frac{1}{2} A I_s \quad \xrightarrow{dP/dT=0}$$

$$(a + T)^2 = 2g_m l a \quad \Rightarrow \quad \boxed{T_m = \sqrt{2g_m l a} - a} \quad \Rightarrow \quad \boxed{P_m = \frac{1}{2} A I_s \left(\sqrt{2g_m l} - \sqrt{a} \right)^2}$$

➤ When $T \ll 1$, $2\delta \approx T + a$, a is the single-loop net loss factor



Discussion 3: Output power $\sim$ length of gain material ($l \uparrow \rightarrow P \uparrow$)

Discussion 4: Output power $\sim$ property of gain material $P = \frac{1}{2} I_s T A \left(\frac{g_m l}{\delta} - 1 \right)$

four-level system $\left\{ \begin{array}{l} P = \frac{1}{2} I_s T A \left(\frac{g_m l}{\delta} - 1 \right) \\ I_s = h\nu_0 / \sigma_{21} \tau_2 \\ g_m = (n W_{03} \tau_2) \sigma_{21} \end{array} \right\} \Rightarrow P = \frac{1}{2} T A h\nu_0 \left(n W_{03} \frac{l}{\delta} - \frac{1}{\sigma_{21} \tau_2} \right)$

$W_{03} \uparrow, n \uparrow, \sigma_{21} \uparrow, \tau_2 \uparrow \rightarrow P \uparrow$

three-level system $\left\{ \begin{array}{l} I_s = \frac{h\nu_0}{2\tau_{21}} \left(\frac{1}{\tau_2} + W_{13} \right) \\ g_m = \frac{n(W_{13} - 1/\tau_2)}{W_{13} + 1/\tau_2} \sigma_{21} \\ P = \frac{1}{4} A T h\nu_0 \left[W_{13} \left(\frac{nl}{\delta} - \frac{1}{2\sigma_{21}} \right) - \frac{1}{\tau_2} \left(n \frac{l}{\delta} + \frac{1}{\sigma_{21} \tau_2} \right) \right] \\ W_{13} \uparrow, n \uparrow, \sigma_{21} \uparrow, \tau_2 \uparrow \rightarrow P \uparrow \end{array} \right.$

$\Delta n_0 = \frac{n(W_{13} - 1/\tau_2)}{W_{13} + 1/\tau_2}$

$\tau_2 = \frac{1}{A_{21} + S_{21}}$

$W_{13} > 1/\tau_2$
to have $\Delta n^0 > 0$

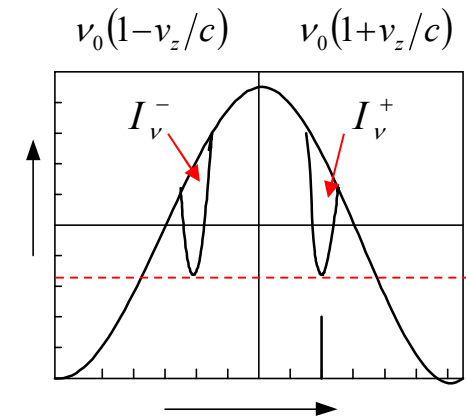
Note for this physical model 1) assume $T \ll 1$; 2) $I_v^+ = I_v^-$

□ Lasers with inhomogeneous broaden (single-mode)

When $\nu_q \neq \nu_0$, I_ν^+ and I_ν^- will have their own burning-holes

$$g_i(\nu_q, I_{\nu_q}) = \frac{g_i^0(\nu_q)}{\sqrt{1 + \left(\frac{I_\nu^+}{I_s}\right)}} = \frac{\delta}{l} \Rightarrow I_\nu^+ = I_s \left[\left(\frac{g_i^0(\nu_q)l}{\delta} \right)^2 - 1 \right]$$

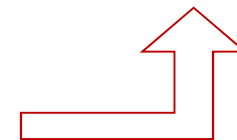
$$\Rightarrow I_\nu^+ = I_s \left\{ \left[\frac{g_m l}{\delta} \exp \left(-4 \ln 2 \frac{(\nu_q - \nu_0)^2}{\Delta \nu_D^2} \right) \right]^2 - 1 \right\}$$



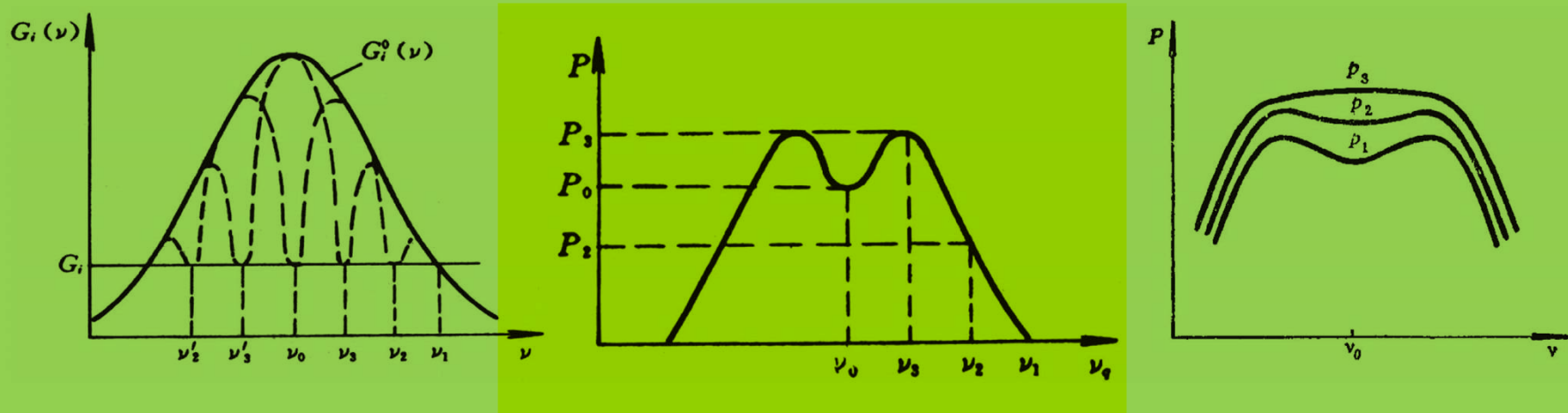
$$P = AI_\nu^+ T = AI_s T \left[\left(\frac{g_i^0(\nu_q)l}{\delta} \right)^2 - 1 \right]$$

$$P = AI_\nu^+ T = \frac{1}{2} AI_s T \left[\left(\frac{g_m l}{\delta} \right)^2 - 1 \right]$$

When $\nu_q = \nu_0$ $I_\nu = I_{\nu_0}^+ + I_{\nu_0}^- \approx 2I_{\nu_0}^+$ similarly we have



Lamb Dip: Relation between single mode output power and frequency



➤ $P \propto$ area of the burning-hole (characterize the inverted populations which contribute to the lasing)

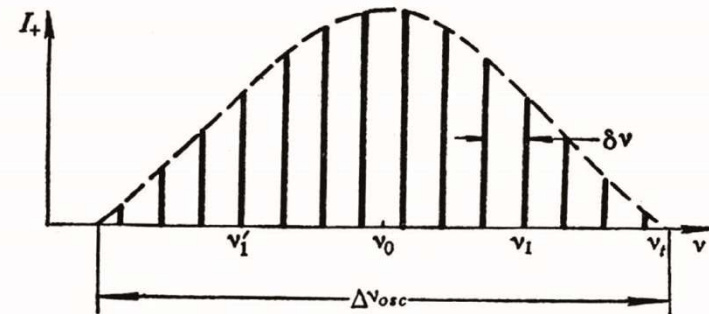
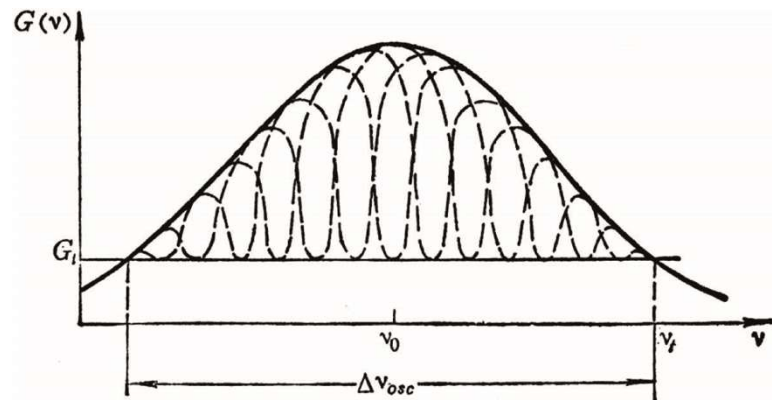
- Burning-hole overlap condition $|\nu_q - \nu_0| < \frac{\Delta\nu}{2} \sqrt{1 + \frac{I_{\nu_q}}{I_s}}$
- Width of the Lamb dip (δ_ν) $\approx$ Hole width $\delta\nu = \Delta\nu_H \sqrt{1 + \frac{I_{\nu_q}}{I_s}}$
- Width of the Lamb dip (δ_ν) $\sim \Delta\nu_L$

Pressure $\uparrow \rightarrow$ Collision broadening $\Delta\nu_L \uparrow \rightarrow$ Lamb dip width $\delta_\nu \uparrow$, shallow depth

❑ Multi-mode lasers

➤ Inhomogeneous broadening:

$\Delta\nu_q$ large enough, NO overlapping of burning holes, calculate the output power for each longitudinal mode, their sum will be the total output power.



➤ Homogeneous broadening: (solid-state lasers)

P_{out} must be calculated using the multi-mode rate equation with simplified assumption (same loss for each mode, rectangular linear function)

$$P = \frac{\overset{\text{Area of the laser beam's cross-section}}{\cancel{A}} \nu_0}{\underset{\text{Area of the gain material's cross-section}}{\cancel{S}} \nu_p} \eta_0 \eta_1 P_{pt} \left(\frac{P_p}{P_{pt}} - 1 \right)$$

❖ Output power of short-pulse lasers ($t_0 \ll \tau_s$)

Four-level system:

output energy $\rightarrow$ stimulated photon number $\rightarrow n_2 \rightarrow$ absorbed pumping energy

$E_0 \rightarrow E_3$		$E_3 \rightarrow E_2$		$E_2 \rightarrow E_1$		E_{in}
Absorb		Activated		Stimulated		Intra-cavity
pumping	$\rightarrow$	photon	$\rightarrow$	photon	$\rightarrow$	laser
energy		number		number		energy
E_p		$\frac{E_p \eta_1}{h\nu_p} > \Delta n_t V$		$\left(\frac{E_p \eta_1}{h\nu_p} - n_{2t} V \right)$		$h\nu_0 \left(\frac{E_p \eta_1}{h\nu_p} - n_{2t} V \right) \frac{A}{S}$
		η_1 – Pumping efficiency				
		V – Volume				$E_{pt} = \frac{h\nu_p \Delta n_t V}{\eta_1}$

$\rightarrow$ Output energy (E)

$$\frac{A \nu_0}{S \nu_p} \frac{T}{T+a} \eta_1 E_{pt} \left(\frac{E_p}{E_{pt}} - 1 \right) \xrightarrow{\frac{T}{T+a} = \eta_0} \boxed{E = \frac{A \nu_0}{S \nu_p} \eta_0 \eta_1 E_{pt} \left(\frac{E_p}{E_{pt}} - 1 \right)}$$